Disconnected qTMD/PDF One-Point Building Blocks

Stochastic Quark Loops for Pion and Proton qTMD Analysis

PyQUDA Measurement Notes

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Contents

1 Purpose

This note documents the PyQUDA implementation of disconnected qTMD/PDF one-point quark loops. The source file is `pyquda_measurement_utils/Disconnected_1pt_qTMD_vibe_develop.py`, and the Perlmutter application is `application/qTMD_disconnected_1pt/perlmutter`.

The loops are hadron independent. Pion and proton information enters later, when the loop is combined with the corresponding hadron two-point function in ensemble analysis.

2 Disconnected qTMD Loop

For a stochastic source ξ_r, define

$$\eta_r = D^{-1}\xi_r, \tag{1}$$

where D is the same clover Dirac operator used by the connected qTMD measurements. Let $P_{q,\tau}$ project the left endpoint onto time τ and multiply it by $e^{i\mathbf{q}\cdot\mathbf{x}}$. For the bilocal operator

$$\overline{q}(x)\,\Gamma\,U_{\mathcal{C}}(x, x+b)\,q(x+b), \tag{2}$$

define $O_b f(x) = U_{\mathcal{C}}(x, x+b) f(x+b)$. The required loop building block is

$$L_{\Gamma,b}(q,\tau) = \mathrm{Tr}\left[P_{q,\tau}\Gamma O_b D^{-1}\right]. \tag{3}$$

Its stochastic estimator is

$$\widehat{L}^{(r)}_{\Gamma,b}(q,\tau) = \xi_r^\dagger P_{q,\tau}\Gamma O_b \eta_r = \sum_{\mathbf{x}} e^{i\mathbf{q}\cdot\mathbf{x}} \xi_r^\dagger(x)\Gamma(O_b\eta_r)(x). \tag{4}$$

Indeed, with the combined site–spin–color indices a, b,

$$\mathbb{E}_\xi[\xi_a^* \xi_b] = \delta_{ab}, \tag{5}$$

$$\mathbb{E}_\xi\left[\xi^\dagger A \xi\right] = \sum_{a,b} A_{ab}\delta_{ab} = \mathrm{Tr}\,A, \qquad A = P_{q,\tau}\Gamma O_b D^{-1}. \tag{6}$$

Thus O_b acts on the solved field $\eta = D^{-1}\xi$, while the original noise remains at the left endpoint. The projector fixes the insertion time; the loop is a spatial trace at each stored τ, not a trace over time.

2.1 Removed reversed estimator

The removed implementation instead contracted

$$\eta^\dagger P_{q,\tau}\Gamma O_b \xi, \tag{7}$$

whose noise average is

$$\mathrm{Tr}\left[(D^{-1})^\dagger P_{q,\tau}\Gamma O_b\right], \tag{8}$$

and therefore is not the required trace. Using $(D^{-1})^\dagger = \gamma_5 D^{-1}\gamma_5$, together with the fact that $P_{q,\tau}$ and O_b are spin diagonal, the reversed trace can be written as the target trace with Γ replaced by $\gamma_5\Gamma\gamma_5$. Consequently it leaves the 5, I, and six tensor labels unchanged but reverses the sign of $T/X/Y/Z$ and $T5/X5/Y5/Z5$.

This identity diagnoses the error only. No channel remapping or data migration is supported. All disconnected qTMD shards and canonical files made with the reversed estimator are invalid and must be deleted and regenerated.

3 Supported Operators

The implementation supports four operator kinds:

`operator_kind`	Operator	Status
`GI_PDF`	Straight-z gauge-invariant PDF Wilson line	Implemented
`CG_PDF`	Straight-z coordinate-gauge displacement	Implemented
`CG_qTMD`	qTMD-style coordinate-gauge displacement	Implemented
`GI_qTMD`	Fixed-length gauge-invariant staple	Implemented

3.1 Straight-z PDF Operators

For PDF operators,

$$b = b_z \hat{z}. \tag{9}$$

The coordinate-gauge version is a plain lattice shift:

$$O_b^{\mathrm{CG\_PDF}} \eta(x) = \eta(x + b_z \hat{z}). \tag{10}$$

The gauge-invariant version applies the straight Wilson line through repeated covariant shifts:

$$O_b^{\mathrm{GI\_PDF}} \eta(x) = U_z(x) U_z(x + \hat{z}) \cdots \eta(x + b_z \hat{z}), \tag{11}$$

with the corresponding backward-link product for negative b_z. In code this is implemented by repeated calls to `gauge.pure_gauge.covDev`.

3.2 Coordinate-Gauge qTMD Operator

The current qTMD operator follows the connected qTMD diagnostic convention:

$$O_b^{\mathrm{CG\_qTMD}} \eta(x) = \eta(x + b_T \hat{e}_\perp + b_z \hat{z}), \tag{12}$$

where $\hat{e}_\perp$ is either $\hat{x}$ or $\hat{y}$. The two transverse directions are stored as `b_X` and `b_Y` in the HDF5 output.

3.3 Gauge-Invariant qTMD Staple Operator

The production code also supports a fixed-length gauge-invariant staple operator, `operator_kind=GI_qTMD`. Its Wilson index is $[b_T, b_z, \eta, \perp]$, with even physical b_z, non-negative η, and $\eta \geq |b_z|/2$.

4 Momentum Convention

The loop momentum is the current momentum transfer,

$$q = p_f - p_i. \tag{13}$$

The one-point loop itself has no hadron source or sink momentum. Those labels enter when the loop is combined with a pion or proton two-point function.

The code uses `pyquda_utils.phase.MomentumPhase`, whose convention is

$$\Phi_q(x; x_0) = \exp\left[+2\pi i \sum_{j=x,y,z} \frac{q_j(x_j - x_{0,j})}{L_j}\right]. \tag{14}$$

The disconnected loop is projected as

$$L_{\Gamma,b}(q, \tau) = \sum_{\mathbf{x}} \Phi_q(\mathbf{x}; \mathbf{x}_0)\, \ell_{\Gamma,b}(\mathbf{x}, \tau), \tag{15}$$

so a local test field with plane-wave dependence

$$\ell_{\Gamma,b}(\mathbf{x}, \tau) \propto \exp\left[-2\pi i \sum_{j=x,y,z} \frac{q_j(x_j - x_{0,j})}{L_j}\right] \tag{16}$$

is selected by the $+q$ projection. This is the same Fourier sign used in the connected qTMD/EMT application layers; the physical transfer label remains $q = p_f - p_i$.

5 Combination with Hadron Two-Point Functions

For a hadron two-point correlator $C_2^H(p_f, t)$, the disconnected qTMD three-point building block is formed as

$$C_{3,\Gamma,b}^{H,\mathrm{disc}}(p_f, p_i; t, \tau) = \left\langle C_2^H(p_f, t)\, L_{\Gamma,b}(q, \tau)\right\rangle_{\mathrm{cfg}} - \left\langle C_2^H(p_f, t)\right\rangle_{\mathrm{cfg}} \left\langle L_{\Gamma,b}(q, \tau)\right\rangle_{\mathrm{cfg}}, \tag{17}$$

with

$$q = p_f - p_i. \tag{18}$$

The corresponding ratio is typically

$$R_{\Gamma,b}^{H,\mathrm{disc}}(t, \tau) = \frac{C_{3,\Gamma,b}^{H,\mathrm{disc}}(p_f, p_i; t, \tau)}{C_2^H(p_f, t)}, \tag{19}$$

before applying the qTMD/PDF renormalization and soft-factor analysis.

The vacuum-subtraction term is essential because the one-point loop is hadron independent.

6 Stochastic Sources and Hierarchical Probing

The source scheme is controlled by

$$\texttt{noise\_scheme} \in \{\texttt{zn}, \texttt{hierarchical\_probing}\}. \tag{20}$$

For hierarchical probing, a base source ξ_b is multiplied by Rademacher probing vectors $h_k(x) = \pm 1$:

$$\xi_{b,k}(x) = h_k(x)\xi_b(x). \tag{21}$$

The number of effective inversions is

$$N_{\mathrm{eff}} = N_{\mathrm{base}}\, N_{\mathrm{HP}}. \tag{22}$$

The implemented HP ordering choices are

Ordering	Meaning
`global_xyzt_gray_projected_to_evenodd`	Gray code the full x, y, z, t site id.
`spatial_xyz_then_t_gray_projected_to_evenodd`	Gray code spatial sites first, then append time bits.

Spin-color dilution and time dilution are not currently implemented.

The default base field is decomposition-independent full-volume $\mathbb{Z}_4$ noise. Its fixed counter contains the global four-dimensional coordinate, spin, color, gauge-configuration number, base-noise index, and an independent stream salt. Therefore any base or HP interval can be regenerated without consuming the earlier random-number sequence. Using the same backend RNG seed independently on equal-shaped MPI sublattices would instead repeat the same rank-local field, violate the required off-diagonal noise covariance, and make the estimator decomposition dependent. Adding the rank to the seed is not sufficient for cross-decomposition reproducibility; the global-coordinate counter is the production solution.

6.1 Base-Oriented Production and Finalization

Production uses only base/HP-interval shards. One complete base estimator is

$$L_b = \frac{1}{N_{\mathrm{HP}}} \sum_{h=0}^{N_{\mathrm{HP}}-1} L_{b,h}. \tag{23}$$

Part files contain at most 64 solves by default. A partial HP part is only a memory and transfer unit, not an independent resume unit. After all parts of one base have been atomically written, rank 0 appends one exact `baseXXXXXX` line to a fingerprinted text sample log. Production resume reads only this log and never probes shard HDF5 files; therefore logged parts may already have been transferred. An unlogged base is recomputed in full.

Measurement jobs do not publish the canonical loop. A separate serial finalizer runs where all parts have been collected. It infers the layout from base 0, checks each expected part once while streaming it into a temporary canonical HDF5 file, and accumulates the average without retaining the full source axis. Only after successful writeout is the canonical file atomically replaced. Non-overlapping base ranges can be scheduled independently; overlapping jobs are unsupported.

7 HDF5 Layout

New output uses the source-independent helper `get_disconnected_qTMD_loop_file_tag` and canonical name

`qTMD1pt/<lat>.qTMD1pt.<cfg>.<ama>.<sm>.h5`.

The obsolete source-position helper has been removed because the full-volume loop has no hadron source position. New shards and canonical files record `schema_version=3`, `loop_convention=xi_dagger_Gamma_` and `trace_target=Tr[P_qtau Gamma O_b Dinv]`. The finalizer rejects any other convention before replacing a canonical file. It stores:

$$\texttt{raw/loop\_pervec} \quad \text{with shape} \quad [N_{\mathrm{eff}}, N_W, N_\Gamma, N_q, N_t]. \tag{24}$$

The raw bookkeeping datasets are

$$\texttt{raw/base\_noise\_index}, \quad \texttt{raw/hp\_index}. \tag{25}$$

The effective source coordinate is not persisted because it is exactly reconstructed as

$$r = bN_{\mathrm{HP}} + h. \tag{26}$$

The averaged data are divided by the spatial volume and stored in a qTMD-like layout:

$$\texttt{avg/SS/<gamma>/PX...PY...PZ.../b\_X/eta0/bT.../bz....} \tag{27}$$

For y-direction transverse displacements the group is `b_Y`. All serial HDF5 files are opened only by MPI rank zero. Because no fermion flow changes the resident gauge in this workflow, the Dirac gauge is loaded once before the source loop and reused.

8 Validation Strategy

The trace direction is tested without precomputed stochastic HDF5 baselines. A complete basis of source vectors is applied to arbitrary complex matrices D^{-1}, O_b, and Γ, including a fixed-time projector and a nonzero spatial phase. The summed estimator must reproduce

$$\mathrm{Tr}[P_{q,\tau}\Gamma O_b D^{-1}], \tag{28}$$

and must differ from the removed $\mathrm{Tr}[(D^{-1})^\dagger P_{q,\tau}\Gamma O_b]$ for a generic test matrix. A separate field-level test records which object is displaced and requires that only η, never ξ, receives the qTMD shift.

Operator tests generated from the current code cover the local limit, coordinate-gauge positive and negative shifts, cached-link equality with the direct covariant-transport reference, and

$$\texttt{GI\_qTMD}(b_T = 0, \eta = |b_z|/2) = \texttt{GI\_PDF}(b_z). \tag{29}$$

The previous tests that conditionally read old smoke-test HDF5 files were removed because those files contain the reversed estimator and cannot serve as valid regression references.

9 Gauge-Invariant Staple qTMD Design

The `CG_qTMD` operator is only a coordinate-gauge diagnostic. The production `GI_qTMD` operator replaces the plain shift by a staple Wilson line. The convention is the fixed-staple-length convention used by the legacy GPT proton qTMD workflow. In the current notation, b_z is the physical final longitudinal separation and must be an even integer:

$$O^{\mathrm{GI\_qTMD}}_{b,\eta,\perp}\psi(x) = U_{\mathcal{C}(x;b,\eta,\perp)}\psi(x + b_T\hat{e}_\perp + b_z\hat{z}), \tag{30}$$

with

$$b_z \in 2\mathbb{Z}, \qquad \eta \geq \frac{|b_z|}{2}. \tag{31}$$

Production applies this operator with $\psi = \eta = D^{-1}\xi$. The three-segment staple path is

$$\mathcal{C}(x; b, \eta, \perp) : x \to x + \left(\eta + \frac{b_z}{2}\right)\hat{z} \to x + \left(\eta + \frac{b_z}{2}\right)\hat{z} + b_T\hat{e}_\perp \to x + b_z\hat{z} + b_T\hat{e}_\perp. \tag{32}$$

The signed segment lengths are

$$\left[\left(z, \eta + \frac{b_z}{2}\right), (\perp, b_T), \left(z, \frac{b_z}{2} - \eta\right)\right]. \tag{33}$$

For $\eta \geq |b_z|/2$, the geometric Wilson-line length is independent of b_z:

$$\left|\eta + \frac{b_z}{2}\right| + b_T + \left|\eta - \frac{b_z}{2}\right| = 2\eta + b_T. \tag{34}$$

This fixed length is useful for renormalization because Wilson-line self-energy effects are matched across different b_z at fixed η and b_T. With $b_T = 0$ and $\eta = |b_z|/2$, the staple reduces to the straight `GI_PDF` Wilson line. At $b_T = b_z = \eta = 0$, it reduces to the local operator.

The legacy GPT implementation used an equivalent convention with b_z^{legacy} denoting half of the final longitudinal separation:

$$b_z^{\mathrm{current}} = 2b_z^{\mathrm{legacy}}. \tag{35}$$

Its path comments and implementation correspond to $(\eta + b_z^{\mathrm{legacy}}) + b_T - (\eta - b_z^{\mathrm{legacy}})$, which is the same fixed-length staple after this relabeling.

The `GI_qTMD` implementation keeps the existing `CG_qTMD`, `CG_PDF`, and `GI_PDF` outputs unchanged. The first code-side helper is `gi_qtmd_staple_segments(W_index)`, which maps $[b_T, b_z, \eta, \perp]$ to the signed path segments

$$\left[\left(z, \eta + \frac{b_z}{2}\right), (\perp, b_T), \left(z, \frac{b_z}{2} - \eta\right)\right]. \tag{36}$$

The helper is covered by `tests/test_disconnected_gi_qtmd_staple_design.py`; it fixes the local, straight-PDF, fixed-length generic path, and invalid-index conventions.

The test-only reference helper is `tests/qtmd_gi_reference.py:create_fermion_TMD_GI`. It applies the full staple path from the original local fermion field for each W-index, using repeated `gauge.pure_gauge.covDev` calls.

Production has only the link-cache path. It first builds gauge-only staple transporters with `build_gi_qtmd_staple_links`. The construction converts an identity color matrix field into color-column fermions, applies the same `covDev` path, and converts the result back to a link field. This keeps the transporter convention identical to the reference direct-covariant derivative path while allowing all stochastic sources on the same gauge configuration to reuse the same staple links. Direct source-by-source transport remains test/reference code and is not a runtime mode.

The link-cache implementation is tested by:

1. identity-gauge equality with coordinate shifts;
2. nontrivial S8T32 test-gauge equality between cached links and direct `covDev`;
3. the PDF limit `GI_qTMD`$(b_T = 0, \eta = |b_z|/2) =$ `GI_PDF`(b_z);
4. a nonzero-transverse-staple HDF5 comparison at $(b_T, b_z, \eta) = (1, 2, 1)$, where `link_cache` and `direct_covdev` agree at the floating-point roundoff level.

The next physics validation should compare the local/PDF limits against the connected pion and proton qTMD workflows after constructing the disconnected ratio with matched two-point functions.